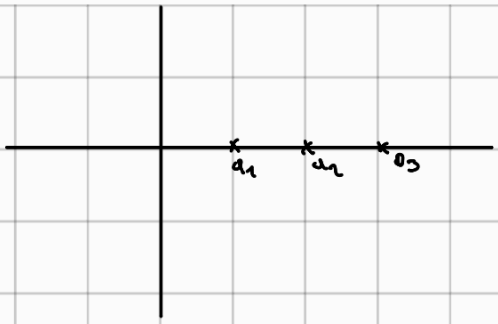


$|z| > a_3$

$a_2 < |z| < a_3$

$a_1 < |z| < a_1$

$|z| < a_1$



$$y[n] = x[n] * h[n]$$

$$Y(z) = X(z)\,H(z)$$

$$H(z) = \frac{Y(z)}{X(z)}$$

$$y[n] = -\sum_{k=1}^{N} a_k\, y[n-k] + \sum_{k=0}^{M} b_k\, x[n-k]$$

$$Y(z) = -\sum_{k=1}^{N} a_k\, z^{-k}\, Y(z) + \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$Y(z)\left[1 + \sum_{k=1}^{N} a_k z^{-k}\right] = X(z)\left[\sum_{k=0}^{M} b_k z^{-k}\right]$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}} \qquad a_0 = 1 \text{ olmalıdır}$$

$$y[n] = \frac{1}{2}\,y[n-1] + 2x[n] - x[n-2]$$

$$Y(z) = \frac{1}{2} z^{-1} Y(z) + 2X(z) - z^{-2} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2 - z^{-1}}{1 - \frac{1}{2} z^{-1}} = 2z^{-1} + 4 + \frac{-2}{1 - \frac{1}{2} z^{-1}}$$

$$h[n] = 2\,\delta[n-1] + 4\,\delta[n] - 2\left(\frac{1}{2}\right)^n u[n]$$

fark denklemi aksi belirtmedikçe nedensel olarak kabul edilir

Ex: $y[n] = 0{,}5\, y[n-1] + x[n]$

$x[n] = 10\cos\left(\frac{\pi}{4} n\right) u[n]$ → sağ yanlı

transient

$y[n] = ?$

$$H(z) = \frac{1}{1 - 0{,}5 z^{-1}} \qquad |z| > \frac{1}{2}$$

$$h[n] = \left(\frac{1}{2}\right)^n u[n]$$

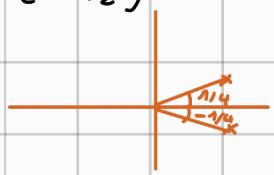
$r^n x[n] \longleftrightarrow$?

$\cos(\omega n)\, u[n] \longleftrightarrow$

$\frac{e^{j\omega n} + e^{-j\omega n}}{2} u[n]$

$$X(z) = \frac{10\left(1 - \frac{1}{\sqrt{2}} z^{-1}\right)}{1 - \sqrt{2} z^{-1} + z^{-2}}$$

$$\left(1 - e^{j\pi/4} z^{-1}\right)\left(1 - e^{-j\pi/4} z^{-1}\right) \rightarrow |z| > 1$$



$\frac{1}{2} e^{j\omega n} u[n] + \frac{1}{2} e^{-j\omega n} u[n]$

$\frac{1}{2} a_1^n u[n] + \frac{1}{2} a_2^n u[n]$

$$Y(z) = H(z) X(z)$$

$$Y(z) = \frac{10\left(1 - \frac{1}{\sqrt{2}} z^{-1}\right)}{(1 - 0{,}5 z^{-1})\left(1 - \sqrt{2} z^{-1} + z^{-2}\right)}$$

$$= \frac{10\left(1 - \frac{1}{\sqrt{2}} z^{-1}\right)}{(1 - 0{,}5 z^{-1})\left(1 - e^{j\pi/4} z^{-1}\right)\left(1 - e^{-j\pi/4} z^{-1}\right)}$$

$$= \frac{A}{1 - 0{,}5 z^{-1}} + \frac{B}{1 - e^{j\pi/4} z^{-1}} + \frac{C}{1 - e^{-j\pi/4} z^{-1}}$$

$$A = Y(z)\left(1 - 0{,}5 z^{-1}\right)\Big|_{z = 1/2}$$

Ara işlemler sınavda

$ax^2 + bx + c$ $\quad \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$y[n] = 6{,}3\,(0{,}5)^n u[n] + 6{,}7\,\delta\, e^{-j\,25{,}7^\circ} \left(e^{j\pi/4}\right)^n u[n]$$

$$+ 6{,}7\, e^{j\,25{,}7^\circ} \left(e^{-j\pi/4}\right)^n u[n]$$

$$= 6{,}3\,(0{,}5)^n u[n] + 13{,}56 \cos\left(\frac{\pi}{4} n - 25{,}7^\circ\right) u[n]$$

transient cevabı

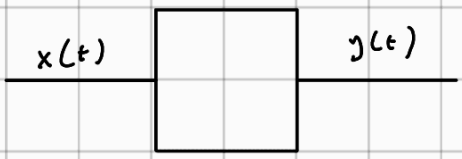
girişe değişmiş hali steady state

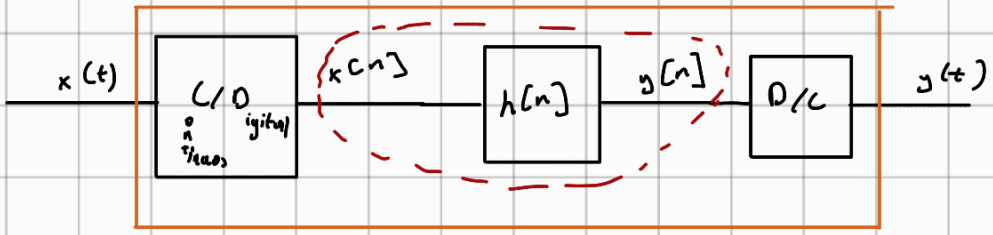
steady state

$$\sum |h[n]| = \sum_{n=-\infty}^{\infty} \left|\left(\frac{1}{2}\right)^n u[n]\right| = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n = \frac{1}{1 - \frac{1}{2}} = 2$$

$$= 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8}$$

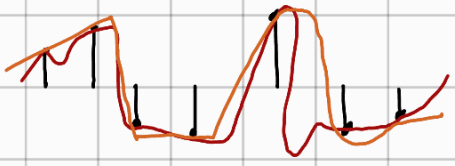
Sampling





$$x(t)\Big|_{t=nT} = x(nT) = x[n]$$



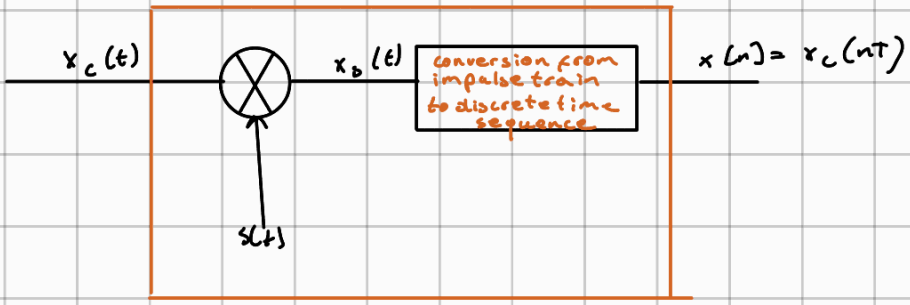


ikisi olabilir

$$x(t) = \cos(12\pi t) \qquad F_s = 24\,Hz$$

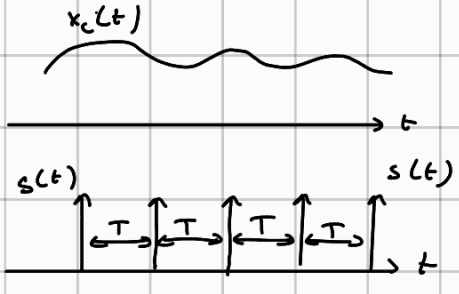
$$x[n] = \cos\left(\frac{1}{2}n\right) \qquad F_s = 6\,Hz$$

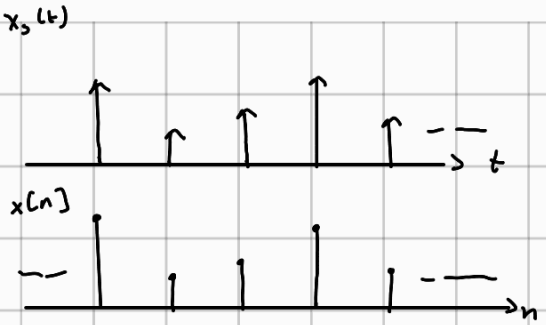
$$x[n] = \cos(2\pi n) = 1$$

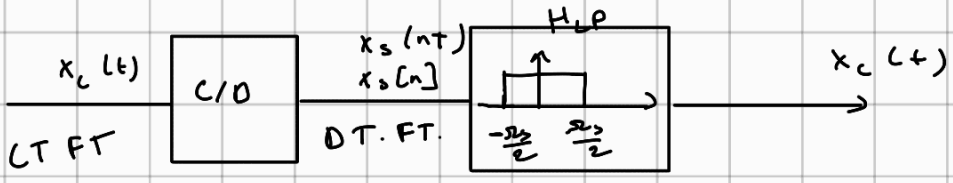


$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$$

$$Y_s(t) = x_c(t) \cdot s(t)$$







$$X_c(j\Omega) = \int_{-\infty}^{\infty} x_c(t)\, e^{-j\Omega t}\, dt$$

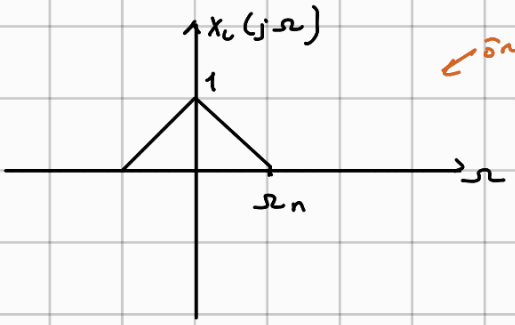
$$X_s(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x_s[n]\, e^{j\omega n}$$

$$X_s(e^{j\omega}) = \frac{1}{T}\sum_k X_c(j(\Omega - k\Omega_s)) \rightarrow \Omega_s = 2\pi f_s,\ T = \frac{1}{f_s}$$

$$= \frac{1}{T}\sum_k X_c\left(\frac{\omega - 2\pi k}{T}\right)$$

olduğunu kabul edelim

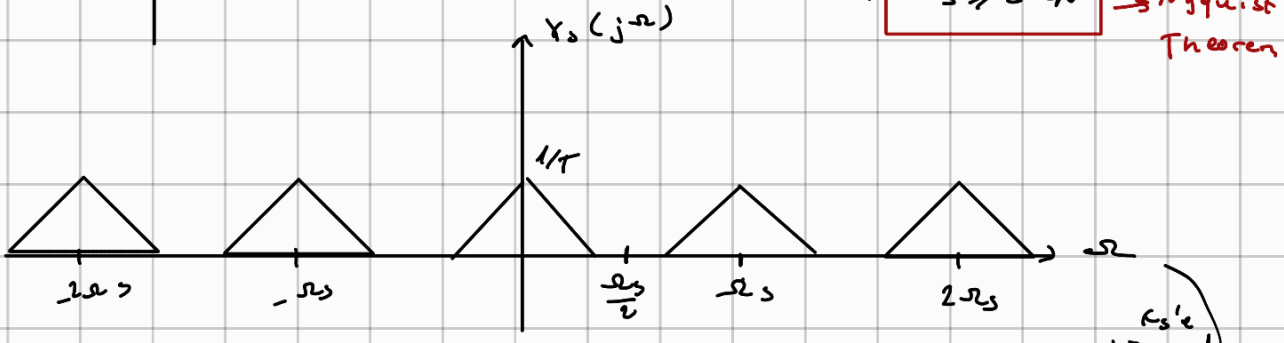
$$X_s(e^{j\omega}) = \frac{1}{T}\Big[\ \cdots + X_c(j(\Omega+2\Omega_s)) + X_c(j(\Omega+\Omega_s))$$
$$+ X_c(j\Omega) + X_c(j(\Omega-\Omega_s)) + X_c(j(\Omega-2\Omega_s))$$
$$+ \cdots\Big]$$

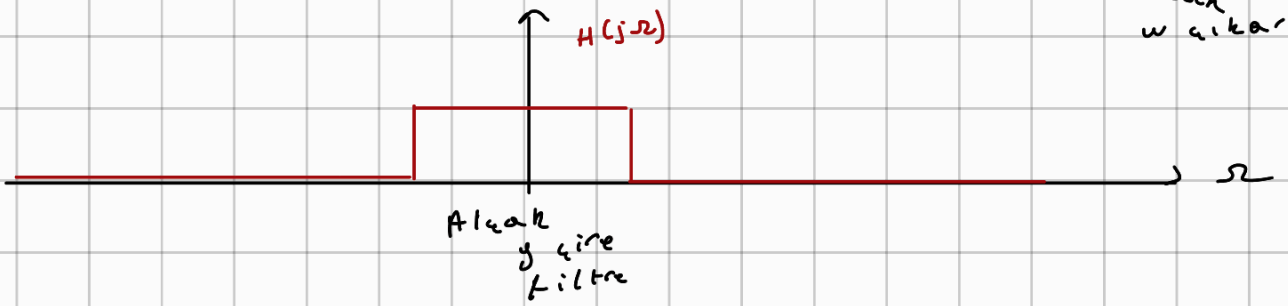


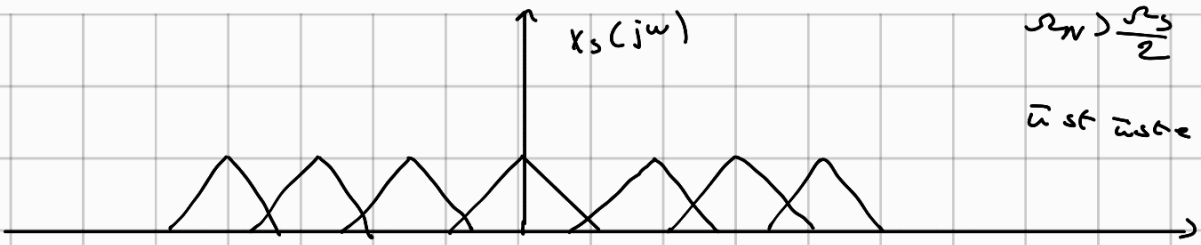
örneklenecek işaretin fourier dönüşümü

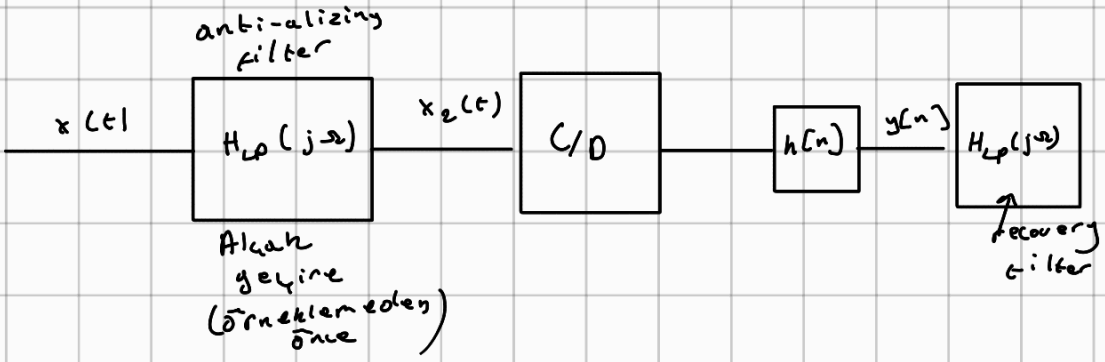
$\Omega_n < \frac{\Omega_s}{2}$

$\rightarrow \boxed{\Omega_s \geq 2\Omega_N}$ → Nyquist Theorem

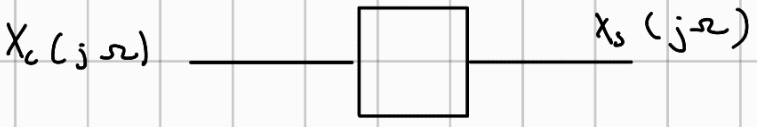








örtüşme aslında iki farklı işaretin örneklerinin aynı olması.



$\Omega_s = \frac{2\pi}{T} = 2\pi B$

$x_s(t) = x_c(t)\, s(t)$ $\qquad s(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT)$

$X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega) \rightarrow S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X_c(j\theta)\, S(j(\Omega-\theta))\, d\theta$ $\qquad x(t) * h(t) = \int x(z)\, h(t-z)\, dz$

$= \frac{1}{2\pi} \int X_c(j\theta) \cdot \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(w-\theta-k\Omega_s)\, d\theta$

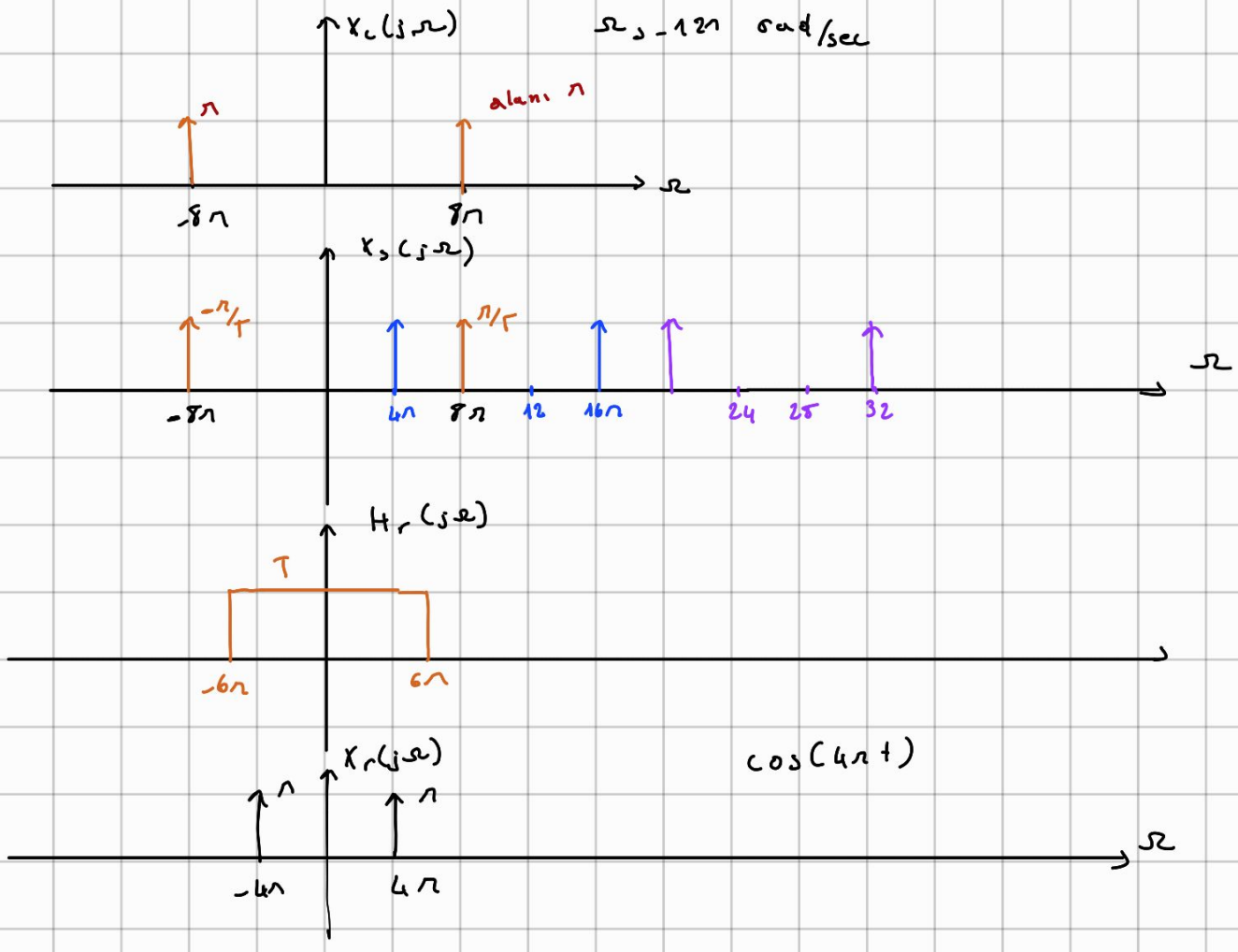
$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$

$w = \frac{\Omega}{F_s} \Rightarrow \Omega = w \cdot F_s \Rightarrow \Omega = \frac{w}{T}$

$X_s(jw) = \frac{1}{T} \sum_k X_c\left(j\left(\frac{w-2\pi k}{T}\right)\right)$

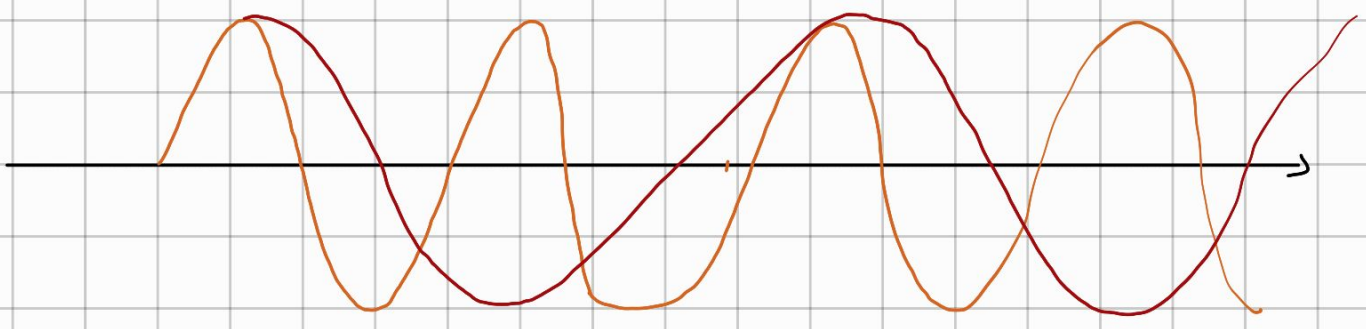
$\frac{2\pi}{T}$

$x_c(t) = \cos(8\pi t)$ $\quad F_s = 6\,Hz$ $\quad \Omega_s = 12\pi$ rad/sec



$\cos(4\pi t)$

örtüşmeyen $\cos(4\pi t)$ ve $\cos(8\pi t)$ örnekleri aynı oldu



$$\cos(8\pi t)\Big|_{t\,en\,\frac{1}{6}} = \cos\left(\frac{4\pi}{3}n\right)$$

$$\cos(4\pi t)\Big|_{t\,en\,\frac{1}{6}} = \cos\left(\frac{2\pi}{3}n\right)$$